BEAM-INSTRUMENTATION NOTE · FOR T10 BEAM PHYSICS

Absolute calibration of the T10 XCET Cherenkov counters,
and what it says about the beam

A user-campaign gift-back: the two threshold counters, calibrated in photoelectrons — plus the beam's measured electron content and a ready-to-use operating table.

Campaign: RADiCAL, PS East Area T10, 25–31 Aug 2026

Counters: XCET040, XCET043 (both upstream of the module)

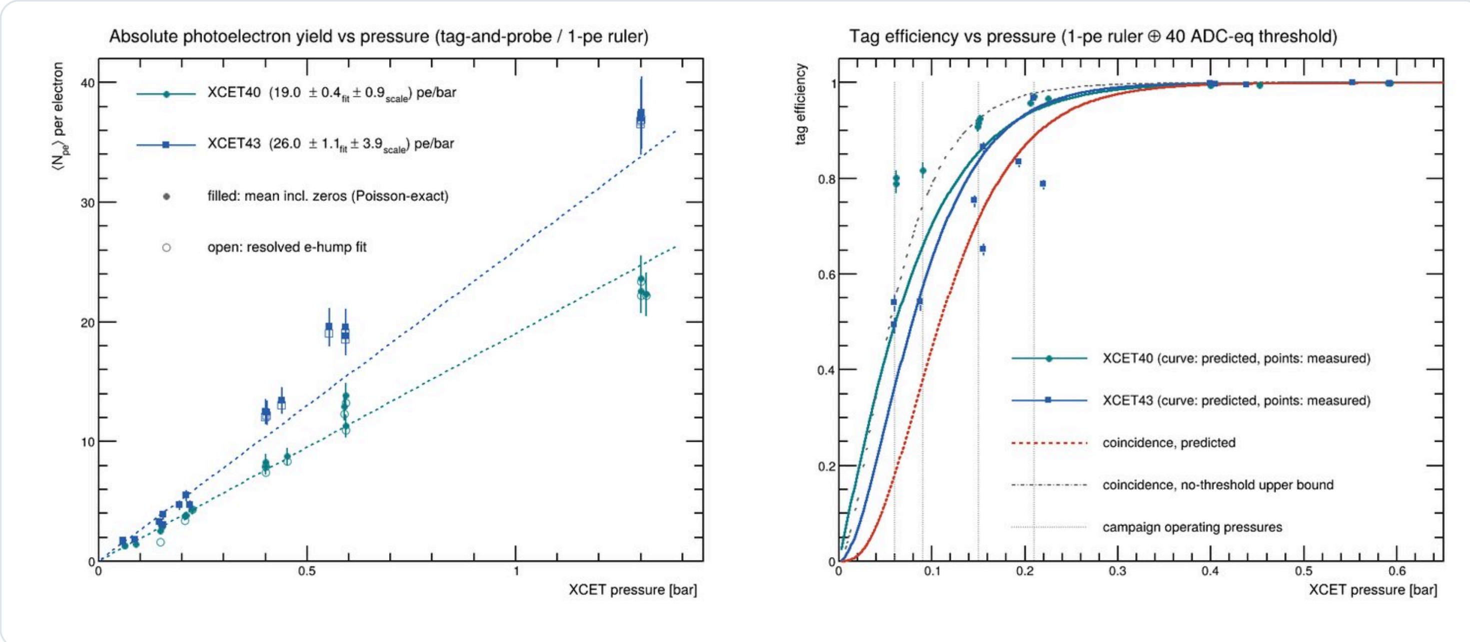
Author: J. Wetzel (RADiCAL) · james@animal-lamps.com **Data:** 18 calibration runs, MCP-triggered

Over five days on T10 we ran the two XCET threshold Cherenkov counters across 18 pressure/energy points with the DRS4 recording their full waveforms on every MCP trigger. Because the DAQ triggers on the MCP and *not* on the XCETs, both counters are unbiased spectators of every event — which let us calibrate each one **absolutely, in photoelectrons**, using its own single-photoelectron peak as the ruler. This note hands that calibration back to the beamline, together with three things it makes possible: a counter-health readout, an electron-content measurement of the beam itself, and an operating table that trades efficiency against purity.

1 The counters, in photoelectrons

19.0 pe/bar XCET040 photoelectron yield ±0.4 fit ±0.9 scale	**26.0** pe/bar XCET043 photoelectron yield ±1.1 fit ±3.9 scale	**~72** /cm XCET040 Cherenkov figure of merit N_0	**~100** /cm XCET043 N_0 (thinner windows, 4.2-bar vessel)

Yields are linear in pressure over the whole 0.06–1.3 bar range explored. Converting through $N_{pe} = N_0 \cdot L \cdot 2(n-1)P$ with $L \approx 3$ m and $(n-1)_{CO_2} \approx 4.4\times10^{-4}$/bar gives the figures of merit above — both inside the canonical 60–100/cm band for a clean gas threshold counter, with the lower-pressure XCET043 vessel coming out higher, as its thinner windows would predict. The single-photoelectron rulers are **40.2 ADC-eq** (XCET040, stable to ±5% RMS over 15 runs) and **~21.8 ADC-eq** (XCET043; its PMT ran at roughly half XCET040's gain, so its 1-pe sits on the pedestal edge and carries a ±15% absolute-scale systematic — see §6).



Left: tag-and-probe photoelectron yield vs pressure for both counters (filled: Poisson-exact mean estimator; open: resolved-hump fits). **Right:** threshold-folded tag efficiency vs pressure, with measured tag-and-probe points overlaid and the Poisson no-threshold bound shown for reference. Run 34 (a 1.3-bar tag at +5 GeV, which radiates on pions) is excluded from the fits.

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2 Counter-health readout

Everything here is measured from the same waveforms — monitoring data that is otherwise awkward to get.

QUANTITY	XCET040	XCET043	HOW MEASURED
Dark occupancy > 40 ADC-eq	1.7% / window	0.4% / window	off-time, pileup-subtracted
Dark occupancy > 60 ADC-eq	0.48%	<0.05%	off-time
1-pe gain stability	±5% RMS / 3 d	single-run	per-run 1-pe fits
Acceptance match (see same beam)	≤0.1% mismatch (time-matched windows)		tag–probe miss floor
In-window pileup rate	~0.13% / event		common-mode (both counters)

A practical note for XCET040: its 1-pe amplitude (40.2 ADC-eq) sits *exactly* at a natural 40-ADC-eq tag threshold, so a single dark photoelectron can fire a tag. That is a ±15%-of-gain coincidence with our threshold choice, not a counter fault — but it is worth knowing before setting discriminators (see §4, §6).

3 Operating table — efficiency vs purity · the one to keep

The electron tag efficiency depends only on pressure (electrons always radiate). **Purity** is the momentum-dependent constraint: below the muon Cherenkov threshold $P_\mu = 12.96/p^2$ bar the tag is pure electrons; above it, muons tag too (and above $P_\pi = 1.745\,P_\mu$, pions as well — the run-34 lesson). The two constraints define the usable window.

|p| [GeV/c]	P FOR 90% COINC.	P FOR 95%	P FOR 99%	μ thr. P_μ	π thr. P_π	MAX *PURE-E* COINC. EFF.
±1	0.22	0.26	0.35	12.960	22.615	—
±3	0.22	0.26	0.35	1.440	2.513	—
±5	0.22	0.26	0.35	0.518	0.905	—
±7	0.22	0.26	0.35	0.264	0.462	95%
±9	0.22	0.26	0.35	0.160	0.279	74%
±11	0.22	0.26	0.35	0.107	0.187	48%

The operational punchline. At 1–5 GeV/c you can reach **99% pure-electron** coincidence tagging — the pressure you need (0.35 bar) sits far below the muon threshold. At 7 GeV/c, 95% is still clean but 99% would start tagging muons. At **9 GeV/c the pure ceiling is ~74%**, and at **11 GeV/c only ~48%**: there is no pressure that is simultaneously high-efficiency and muon-free. This is exactly why our own −11 GeV runs saw the tag deficits they did — the physics was in the counters all along, and this table makes it choosable in advance.

4 What the beam is made of · a cross-check for your model

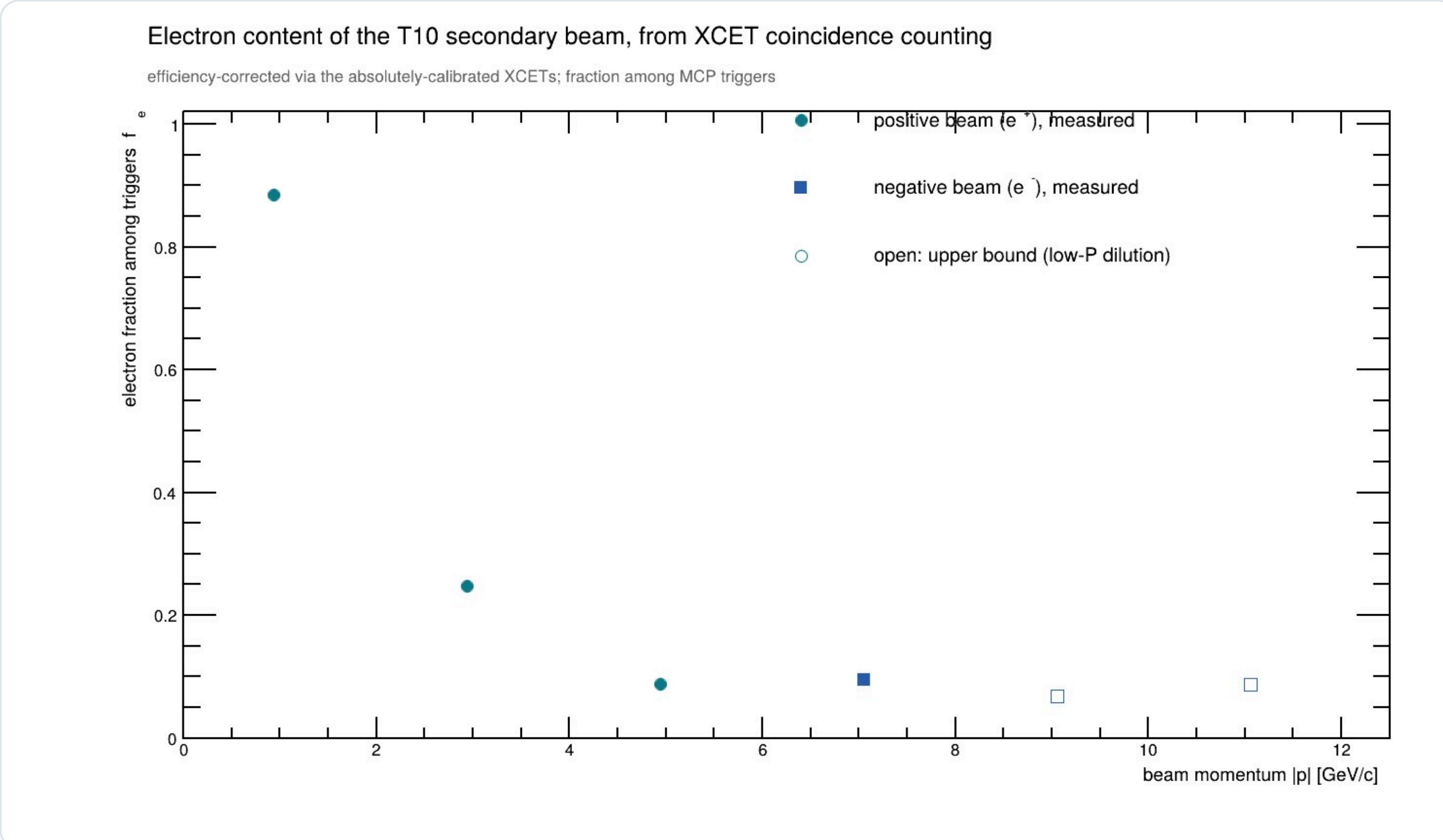
Turning each run's coincidence-tag rate around through the calibrated efficiency gives the beam's **absolute electron fraction** — a direct, independent check of the T10 secondary-beam composition.

88% e^+
positive beam, +1 GeV/c

25%
positive beam, +3 GeV/c

9%
positive beam, +5 GeV/c

~7–9% e^-
negative beam, 7–11 GeV/c



Measured electron fraction among triggers vs |p|. The positive beam is strongly electron- (positron-) dominated at low momentum — **88%** at +1 GeV/c — and falls steeply to ~9% by +5 GeV/c, the expected signature of e^+ from $\pi^0 \to \gamma \to e^+e^-$ conversions in the production target dwindling as the hadron fraction rises. The negative beam is lower and flatter (~7–9%). Open markers are upper bounds: at ≤0.15 bar the tag efficiency is depressed by fake-tag dilution (§6), so the correction over-counts electrons there. Fraction is *among MCP triggers*, not raw spill composition.

One observation worth your eye: at +5 GeV/c the three runs disagree beyond statistics (run 15, Aug 28: 11.8%; runs 37/41, Aug 30: 7.7%) — a genuine day-to-day composition shift, plausibly a beam-tune or collimator change between the two dates.

5 Beam observations from the run

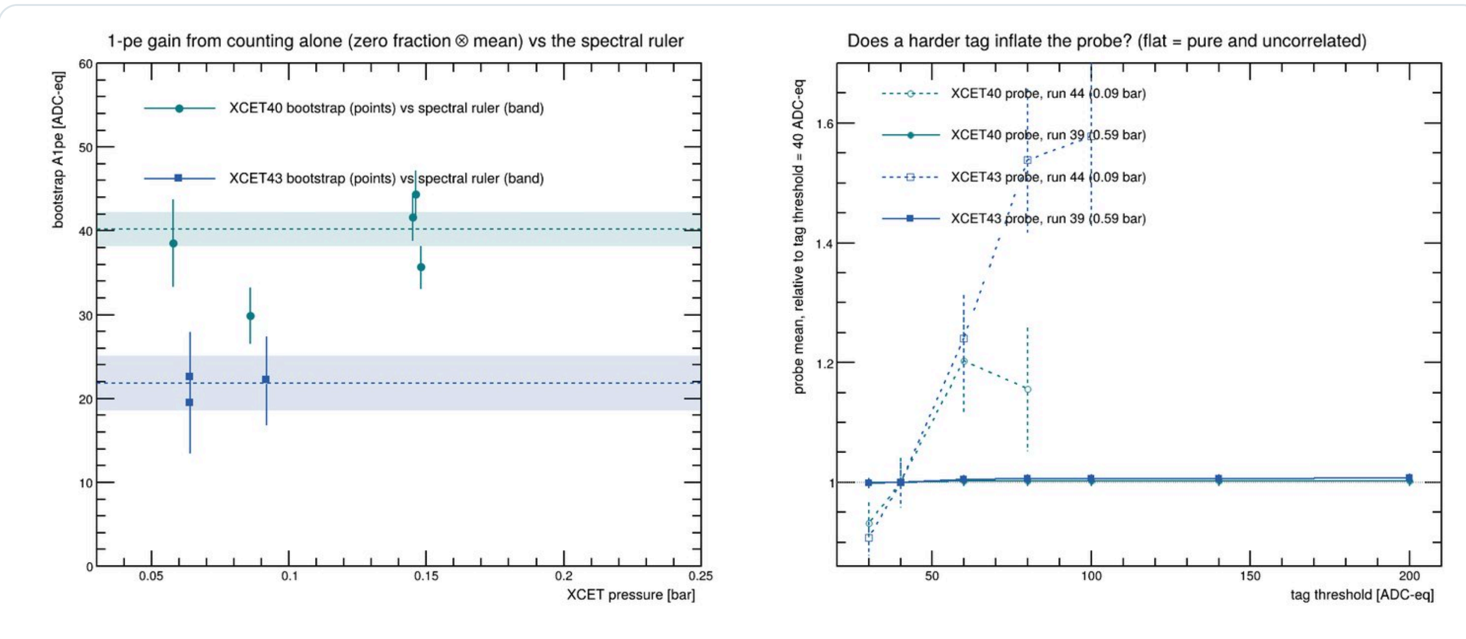
- **Rate recovery at 1 GeV.** The beam arrived very wide and flat in both planes; peak counts on the CESAR profile were ~80 where ~800 was expected. Opening the collimators (from –10/–10/–3/–20.1) and dropping XCET040 from 11 bar recovered the rate substantially — a useful data point on the low-momentum tune.
- **Trigger rate:** ~2500 MCP triggers in 1h15m at 1 GeV before the retune.
- **Pion contamination is real and sharp:** at +5 GeV/c a 1.3-bar tag (run 34) put pions in the electron sample — its probe mean fell 2.6× — exactly as P_π = 0.90 bar at 5 GeV/c predicts. A worked example of the §3 table.

6 Method, briefly — and two things you may not have seen

The calibration is **tag-and-probe**: to measure one counter without biasing it, we select on the *other* counter and read the first with no cut, zeros included. The mean amplitude including zeros, divided by the 1-pe ruler, is the Poisson mean N_{pe} directly — valid even at 0.06 bar where the electron hump dissolves into counting. We validated every assumption (purity, independence, acceptance) on the data itself; two pieces may be new:

> **A gain ruler from counting alone.** On tagged electrons the zero fraction and the mean are two equations in two unknowns (λ, A_{1pe}) — so the single-photoelectron gain can be recovered with *no resolved 1-pe peak and no pedestal fit*. It confirms XCET043's pedestal-edge ruler independently: 21.6 ± 2.3 ± 3.7 ADC-eq against the spectral 21.8.

> **Dark rates without a dark run.** Dark pulses are uniform across the 205-ns window while beam signals are prompt, so an off-time window measures the dark rate on ordinary beam events — and the in-window pileup component (which lights *both* counters) is separated as the common-mode rate above the single-counter dark reach. No dedicated pedestal run required.



Left: the counting bootstrap — 1-pe gain recovered from beam electrons alone (points) against the spectral rulers (bands). **Right:** the fake-tag signature — probe mean vs tag threshold, flat where the tag is pure (0.59 bar) and climbing where dark-photoelectron fakes dilute a scarce electron sample (0.09 bar).

The one caveat that rides everywhere: at low pressure (≤0.15 bar) fake tags — a dark photoelectron in the tag counter over a non-radiating hadron — dilute the probe by a measured 15–40% (XCET043) / 4–14% (XCET040). It biases the low-pressure efficiency points and the §4 open markers low; it is quantified, not merely flagged, in the methods note.

7 Open questions — where your knowledge would help

- **Vessel & optics geometry.** We assumed L ≈ 3 m for both. The genuine 35% N_0 difference we see (XCET043 > XCET040) we attribute to the 4.2-bar vessel's thinner windows — can you confirm the radiator lengths and window materials?
- **Pressure-gauge provenance & zero.** Are the registry pressures CESAR readbacks or local transducers, and what is the zero offset at the low end? At 0.06 bar a ±0.01 bar offset is a 16% effect on our lowest points.
- **A mid-pressure amplitude correlation** between the two counters (Pearson r up to 0.68 near 0.44 bar, run-dependent) is not explained by the measured pileup rate alone — any known intensity or geometry effect?

8 What we can add if useful

> **To finalize this as an EDMS/ELOG note, we would fold in from your side:** the campaign HV settings and PMT serials for both counters; the CO_2 fill/purge history (did the run-36-era gas intervention touch either counter?); hall temperature during the calibration (pe/bar is really pe per gas density — a 10 K swing is ~3%); and the pressure-readback provenance above. With those, every number here becomes a dated, reproducible reference.

Two recommendations, evidence attached: (1) next campaign, raise XCET043's HV (or lower the DRS4 range) so its 1-pe peak clears the pedestal — the ±15% scale systematic then collapses to XCET040's ±5%. (2) Avoid setting a tag threshold *at* a counter's 1-pe amplitude (our 40-on-40.2 coincidence) — it cost us 15–40% fake-tag dilution at low pressure.

A Appendix — per-run electron content (traceability)

Every point behind §4. Raw rate = coincidence tags / triggers; ε_{fold} = threshold-folded coincidence efficiency at the run's pressures; f_e = raw / ε. Rows marked * (dimmed) are ≤0.15 bar, where fake-tag dilution makes f_e an upper bound. Merged files are expanded to their source runs. Run 34 (pion-contaminated tag) excluded throughout.

RUN	p [GeV/c]	P_{40}/P_{43} [bar]	COINC / TRIG	RAW RATE	ε_{fold}	f_e
15	+5	0.400/0.400	2,347/20,000	11.7%	99.6%	**11.8%**
24	+3	1.300/1.300	5,014/20,000	25.1%	100.0%	**25.1%**
2526→25+26	+1	1.300/1.300	8,839/10,264	86.1%	100.0%	**86.1%**
27	-7	0.210/0.210	1,754/20,000	8.8%	89.0%	**9.9%**
9001→28+29	-9	0.150/0.156	941/20,000	4.7%	73.0%	**6.4%** *
30	-11	0.062/0.060	435/20,000	2.2%	18.6%	**11.7%** *
31	-11	0.062/0.060	453/20,000	2.3%	18.6%	**12.2%** *
32	-9	0.149/0.146	958/20,000	4.8%	70.4%	**6.8%** *
33	-7	0.206/0.194	1,721/20,000	8.6%	86.8%	**9.9%**
35	+3	1.313/1.303	4,964/20,000	24.8%	100.0%	**24.8%**
37	+5	0.400/0.405	1,525/20,000	7.6%	99.6%	**7.7%**
38	+1	0.590/0.553	17,801/20,000	89.0%	100.0%	**89.0%**
39	+1	0.593/0.593	17,790/20,000	88.9%	100.0%	**89.0%**
40	+3	0.593/0.593	4,895/20,000	24.5%	100.0%	**24.5%**
41	+5	0.452/0.439	1,534/20,000	7.7%	99.8%	**7.7%**
42	-7	0.225/0.220	1,554/20,000	7.8%	91.0%	**8.5%**
43	-9	0.152/0.156	1,034/20,000	5.2%	73.3%	**7.1%** *
44	-11	0.090/0.088	496/20,000	2.5%	37.1%	**6.7%** *

RADiCAL Collaboration · T10 beam test, August 2026. Full analysis, macros and per-run data provenance: jwwetzel.github.io/radical-t10-2026. Calibration `macros/XCETCalib.C`; validation `macros/XCETTagProbe.C`; beam content `macros/BeamContent.C`. Raw waveform data available on request. Snapshot: this is a 2026 calibration; window transmission and mirror reflectivity drift, so re-verification at the next access is recommended.